Attention-Aware Dual-Stream Network for Multimodal Face Anti-Spoofing

Pengchao Deng, Chenyang Ge, Xin Qiao, Hao Wei, and Yuan Sun

***Abstract*—Since the rapid development of face recognition systems using 3D cameras, the public has demanded great safety regulations for these devices. As a closely related topic, multimodal face anti-spoofing (FAS) has become an indispensable part of face recognition systems. However, existing multimodal FAS tools suffer from performance degradation under external low-lighting conditions and insufficient representation capabilities of fusion features. To address these issues, we present an attention-aware dual-stream fusion method using 3D cameras (i.e., IR+Depth) and considering both fine-grained and global features. Specifically, we introduce a surface normal generator using depth maps to obtain robust and discriminative representations. Then, we leverage the attention mechanism to split each stream into two branches. The first branch explores complementary global information between different modalities, while the second branch captures subtle and local features from each modality. The system regards multimodal FAS as a fine-grained classification problem. Moreover, to ensure that local areas in the image do not overlap and belong to the same class, a joint loss function is developed and proven to further boost the performance of FAS. We extensively evaluate our proposed strategies on various multimodal databases, and the results show that when compared with current state-of-the-art multimodal methods, our framework achieves superior performance.**

***Index Terms*—Face anti-spoofing, surface normal map, attention mechanism, global and local feature fusion.**

I. Introduction

TRADITIONAL biometrics mainly contain contacted authorization methods such as fingerprints, heartbeats, and keystrokes, which may be inconvenient to daily lives. Since human faces have rich identification and discrimination information, they recently become a powerful clue in the biometric field. In addition, this non-contact and fast authorization method has been widely used in financial payment [1], mobile phone unlocking [2], and access control [3], [4] in recent years.

Simultaneously, various malicious attacks have begun to appear to break the face recognition system and violate personal privacy. These malicious attacks, including 2D attacks (e.g., printed papers [5] and replay attacks [6]) and 3D attacks (e.g., 3D models and 3D masks [7]), make face recognition systems more vulnerable. Thus, face anti-spoofing (FAS) plays a critical role in reinforcing face recognition systems.

Many efforts on face anti-spoofing have been proposed in the past decade. In the early research, most works focus on hand-crafted features [8], [9], [10], [11] resisting 2D attacks. Researchers usually exploit image-processing techniques to obtain these hand-crafted features, which are then fed into shallow classifiers [12], [13], [14]. Nevertheless, these traditional methods are limited by inadequate representation. In recent years, to overcome this limitation, deep learning methods [15], [16], [17] have been proposed for FAS. Along with the development of 3D technology, face masks made of different materials have emerged to threaten face recognition systems. Using only a single modality may be insufficient for the detection of sophisticated attacks. To further obtain additional spoofing clues, a few efforts consider depth maps as auxiliary cues or supervision [18], [19], [20], [21] to boost the performance. However, most of the depth maps are pseudo, i.e., face reconstruction [22] is utilized to produce living faces from RGB images and define values of spoofing faces as 0, resulting in underconstrained outcomes [23].

Recently, a few researchers begin to explore complementary features among different modalities for FAS due to the collection of multimodal databases [24], [25], [26], [27]. Based on these databases, numerous works [28], [29], [30], [31], [32], [33] demonstrate great potential in multimodal FAS. However, there are still several challenges. First, these works build complex network structures with more than three inputs, which brings more parameters and time overhead. Second, while most previous works only relied on RGB cameras, researchers [24], [25], [26], [27] verify that IR images show better results than depth maps and RGB images in most multimodal databases, especially when facing 3D attacks. On the one hand, the qualities of RGB images are easily affected by changes in the external environment (e.g., lighting environment and device settings). On the other hand, although depth maps can identify 2D attacks according to their fixed shapes, they are unable to handle 3D attacks because of exhibiting similar shapes to living faces. In contrast, IR images are formed by receiving and recording thermal radiation energy emitted via object. The thermal radiation energy is different between materials of 3D attacks and skins of living faces, which is a vital cue for FAS to distinguish living faces from spoofing attacks. Third, the faces with similar appearances may not have similar scales. Thus, the uncertainty of the

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

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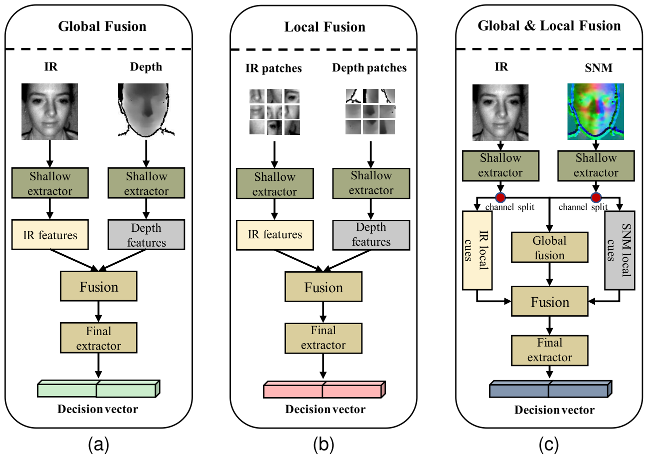


Fig. 1. Compared with existing multimodal strategies. (a) Global Fusion: regards the entire images as inputs from various modalities. (b) Local Fusion: regards various patches as inputs from different modalities. (c) Global & Local Fusion: relies on both global and local fusion information.

scale causes the uncertainty of the depth, which makes it difficult to distinguish the depth between living and spoofing faces accurately. Different from depth, the surface normal map (SNM) is invariant with respect to the scales of objects and can better represent the intra-structure [34]. In addition, changes of expression in depth may lead to the misjudgment of living faces, while SNM has been verified for face recognition to have good robustness to a broad range of expression variations [35], [36]. In recent work [33], SNM has been introduced for FAS and achieves favorable results. However, the algorithm requires accurate point clouds as input, which cannot be used on all multimodal public databases.

Another concern is that existing multimodal approaches fail to sufficiently explore global and local information relationships. Most works [24], [25], [26], [27] adopt a global strategy, which typically fuses complementary information extracted from whole images of different modalities via intermediate feature fusion modules as illustrated in Fig. 1a. However, the differences between living and spoofing faces are sometimes subtle and local [18]. Therefore, exclusively considering global features without subtle and local cues is insufficient to generate discriminative features. Instead of using the entire images as inputs, the local fusion strategy [32], [37], [38] leverages fixed or random patches for subtle and local cues, as shown in Fig. 1b. Spoofing clues, however, can be anywhere (as opposed to being fixed or random), which may result in the loss of important information. Thus, only considering local features is also not enough. The analysis mentioned above encourages us to adopt a local and global fusion strategy, which is depicted in Fig. 1c. Although several previous works [39], [40], [41] have employed neural networks for joint extraction of global and local fusion in different vision tasks, using post-process pretrained global feature models or attention modules in the methods cannot produce more subtle features. Different from these works, we regard multimodal FAS as a fine-grained classification and attempt to employ an attention-aware strategy to boost the performance.

In this paper, we present an attention-aware dual-stream deep framework based on 3D cameras to combine fine-grained and global fusion information. To obtain discriminative and robust intra-structure details of faces, we design a surface normal generator, which is applicable to most public multimodal databases. The normal map and IR are regarded as two-stream inputs to the proposed framework, in which we use attention mechanisms to split each stream into local and global branches. On the one hand, the global branch consists of a Global Interaction Fusion (GIF) block aiming to explore complementary representations from different inputs. On the other hand, the local branch introduces a Local Attention Extraction (LAE) module to focus on subtle and local regions of intermediate feature maps from each input. Furthermore, to enforce local areas in different modalities to belong to the same class and be non-overlapping, we present a joint loss function that includes center loss and cosine similarity loss terms.

To demonstrate the effectiveness of our dual-stream framework, we conduct extensive experiments on different existing multimodal databases. Quantitative and visualized results show that our method achieves remarkable performance in seen and unseen attacks. In summary, the main contributions of this work are four-folds as below:

- We develop an attention-aware dual-stream framework based on 3D cameras, which consider more fine-grained and global fusion features, yielding powerful representations for multimodal FAS. To the best of our knowledge, it is the first time to regard the multimodal FAS as a fine-grained classification task.
- Surface normal maps (SNMs) with a robust discriminative representation between living and spoofing faces are used to align with IR for effective fusion, which is a suitable approach for most public multimodal databases.
- We present a Global Interaction Fusion (GIF) block and Local Attention Extraction (LAE) module for effective global information fusion and local cues capture, respectively. Moreover, a joint loss function is introduced to further boost multimodal FAS performance.
- Comprehensive experiments demonstrate that our proposed approach achieves superior results compared with the state-of-the-art methods on the main multimodal benchmark tests.

II. Related Works

In this section, we review existing FAS methods based on four groups that uniquely consider relevant input: image-based methods, video-based methods, 3D-based methods, and fusion methods. Then we analyze attention-based methods and fine-grained classification works related to our proposed method.

A. Image-Based Methods

Image-based methods identify attacks via different features based on the information, such as RGB images, IR images, and depth maps. These features can be generally divided into hand-crafted features and deep learning-based features.

We first introduce the hand-crafted features. Many previous works use image-based information to extract the hand-crafted features for FAS, including LBP [8], HOG [10], and DOG [11]. These features typically show different decision clues between living and spoofing faces. Moreover, spoofing attacks can be easily launched against face recognition systems on 2D cameras due to the low cost of obtaining printed photos or video replays. The work in [9] proposes an efficient detection approach based on the analysis of image distortions in 2D spoofing face images and the complementarity of individual cues (LBP and color moments), which performs fairly well.

With the development of deep learning strategy [42], more efforts [15], [16], [17], [43] exploit Convolutional Neural Network (CNN) architecture to extract features from image-based information, which show better performance than hand-crafted features. For example, Yang et al. [15] rely on CNNs to learn features of high discriminative ability with binary supervision using RGB images for the first time. Tu et al. [43] combine the multi-task (i.e., face recognition and face anti-spoofing) model with a fast domain adaptation strategy to improve the generalizability and applicability of FAS. However, the discriminative representation ability of these works remains insufficient. To address this issue, many works add other image-based information for FAS. Atoum et al. [18] change RGB images into different patches using CNNs as one stream and input the pseudo depth maps into depth-based CNNs as another stream. They obtain final results by fusing the two-stream scores. Yu et al. [19] design a central difference convolutional network (CDCN) with pseudo depth as discriminative supervision to improve the performance. Peng et al. [44] adopt depth and color (RGB, HSV, YCbCr) information to obtain a more robust representation.

B. Video-Based Methods

Video-based methods typically use temporal cues to obtain more discriminative information in videos for FAS. Motion detection is an early method based on temporal cues, such as eye-blinking [45], [46] and mouth-moving [47]. In [46], their proposed method combines depth maps and motion information of the human face. They show an improved RANSAC to eliminate videos and photo spoofing, along with a new blink detection method to eliminate 3D spoofing. In addition, some methods [48], [49] use different strategies to capture video-specific features. For example, Siddiqui et al. [49] suggest a multi-feature video let aggregation method with multi-LBP and HOOF for FAS.

To extract comprehensive features, several efforts use the recurrent neural network architecture. Xu et al. [50] propose a CNN-LSTM network to capture temporal features. Liu et al. [20] designs a novel CNN-RNN structure with the rPPG supervision using a multi-frame image of RGB and HSV. To capture more helpful features. Zheng et al. [51] introduce a two-stream spatial-temporal network based on depth and multi-scale information. In addition, 3D CNNs are also used to extract temporal features on a continuous video-base frame [52], [53].

C. 3D-Based Methods

3D-based methods fully leverage the 3D spatial signal, which is an important cue to distinguish fixed shape spoofing (2D attacks). There are relatively few methods for 3D information due to the complexity of 3D information processing and extensive storage resources. Two processing strategies can make FAS take advantage of 3D data. One regards fine-grained 3D point clouds as supervision labels to obtain discriminative features, such as [54]. The other is inspired by 3D face recognition methods using normal maps. For example, [55] uses image decomposition to obtain albedo and normal maps for FAS. However, it only emphasizes the importance of albedo maps while neglecting the interpretation of normal maps. In addition, this method cannot get the correct normal map representation of spoofing attacks. Deng et al. [33] generate fitted normal maps using 3D point clouds to obtain discriminative features, which is unsuitable for most public multimodal databases due to the lack of accurate point clouds.

D. Fusion Methods

With the emergence and continuous development of 3D cameras, we can easily access more different modal information at the same time. Researchers leverage different fusion strategies [31], [56], [57] to extract complementary and comprehensive features from multiple modalities. For example, Zhang et al. [24] propose a large-scale multi-modal dataset CASIA-SURF and design a CNN structure to fully use multiple modalities, including RGB images, IR images, and depth maps. George et al. [27] design a pre-trained multi-stream network using their new Wide Multi-Channel Presentation Attack (WMCA) database. Liu et al. [26] introduces a novel multimodal fusion method to alleviate ethnic bias. Yu et al. [28] obtain fusion information from each modality by the CDCN structure, which shows convincing results. Sanghvi et al. [58] build sub-architectures for detecting face presentation attacks to further identify types of attacks.

The abovementioned fusion methods only use image-based information and lack useful video-based temporal and 3D-based information. Videos are composed of images suitable for extracting temporal information rather than fusion information [59]. Since 3D information is discriminative to shapes, we think it is worth fully integrating image-based and 3D-based information into a fused architecture.

E. Attention-Based Methods

Attention-based methods rely on a powerful mechanism that enables perception to focus on important regions providing helpful information for FAS. To fuse temporal features and estimated depth information, [51] employs a scale-level attention module. Chen et al. [60] apply an attention-based fusion method to fuse features derived from RGB images and multi-scale retinex (MSR) images. In addition, [61] introduces two-stream vision transformers with self-attention fusion [62] to obtain complementary information from RGB space and multi-scale retinex with color restoration space. For multimodal FAS, in [63], the authors propose an end-to-end multimodal fusion approach via spatial and channel attention to advance performance. Yu et al. [4] develop a cross-attention fusion module based on the self-attention mechanism to search cross-modal clues for flexible-modal deployment. However,

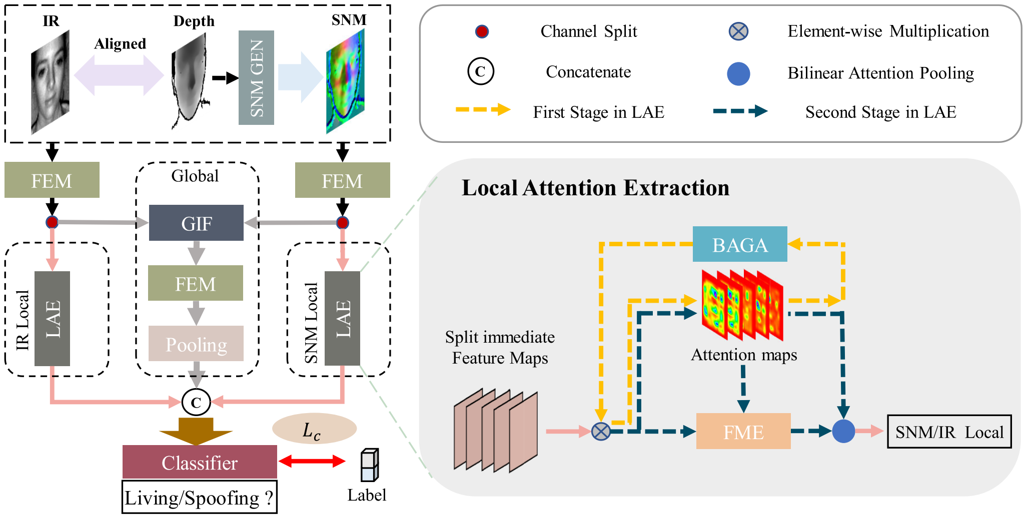


Fig. 2. The detailed architecture of our proposed dual-stream framework. The representation capability of depth maps is enhanced via a surface normal generator (SNM GEN). Local and global cues from different modalities play a role in the final prediction.

these attention-based works concentrate on important parts without distinguishing whether they belong to local or global regions, which may lead to loose some spoofing clues in the fusion process. We think it is useful for multimodal FAS to further subdivide these parts and fuse them separately.

F. Fine-Grained Classification

To effectively overcome the challenge of fine-grained classification, attention-based methods [64], [65] are introduced to distinguish different categories in local regions. Efforts in this field typically focus on locating the discriminative areas and learning subtle cues in weak supervision. Inspired by these works, we regard multimodal FAS as a similar fine-grained classification problem to learning subtle and discriminative features about spoofing attacks. Although multimodal methods [31], [32], [38] using fixed or random patches in FAS have been proposed to focus on local information, these fixed or random patches may result in the loss of important spoofing clues. Different from these methods, we leverage an attention mechanism based on fine-grained classification to capture local features, where attention maps can be learnable according to different inputs.

III. Methodology

In this section, we introduce an attention-aware dual-stream framework that focuses on local and global information fusion for FAS. As illustrated in Fig. 2, our framework comprises two similar streams with different inputs, i.e., depth and IR. First, we capture robust and discriminative information from the depth map using a surface normal generator. Next, intermediate feature maps from normal map and IR are obtained using a Feature Extraction Module (FEM) based on two residual blocks [66] and are divided into two components by channel split operation. Specifically, one component is fed to the global branch which aims to capture global fusion information, while the other serves as input to the local branch, which focuses on subtle spoofing clues. On the one hand, to effectively fuse intermediate feature maps from different global branches, we develop a Global Interaction Fusion (GIF) block motivated by [67]. Following the GIF, the FEM is utilized to capture global semantic information. On the other hand, the local branch is based on a two-stage feedback Local Attention Extraction (LAE) module, which more closely considers the subtle regions. Moreover, the center loss and cosine similarity loss are employed in the LAE modules to provide attention maps with category information and non-overlapping local areas. The final decision vector, at last, can be obtained by a classifier taking as input the local and global fusion cues.

A. Surface Normal Generator

Rather than depth maps, we consider using surface normal maps as input for multimodal methods, which is motivated by depth maps being sensitive to the scales of faces [34] and possibly neglecting local regions that often contain useful clues for FAS. Surface normal maps represent the local curvature of a 3D polygonal mesh in terms of RGB color data, which are widely applied in face recognition systems [35], [36] due to their robust representations. For FAS, only a few studies [33], [55] involve surface normal maps. However, they either cannot represent spoofing attacks or are not applicable to all public multimodal databases. To address these problems, we propose a surface normal generator. Specifically, we first collect a set, including spatial information for each pixel in the depth map M_D with the size of $H \times W$, which is defined as

$$P_{i,j} = \{i, j, z_{(i,j)}\}, P_{i,j} \in P \tag{1}$$

where P is the collected set with the size of $H \times W$, $P_{i,j}$ is a pixel point in P, and i, j are the coordinates of the depth map. $z_{(i,j)}$ is the depth value for each pixel in the depth map, respectively. When obtaining the set, we can calculate the normal according to the definition of the normal vector and principal component analysis (PCA) [33], [68].

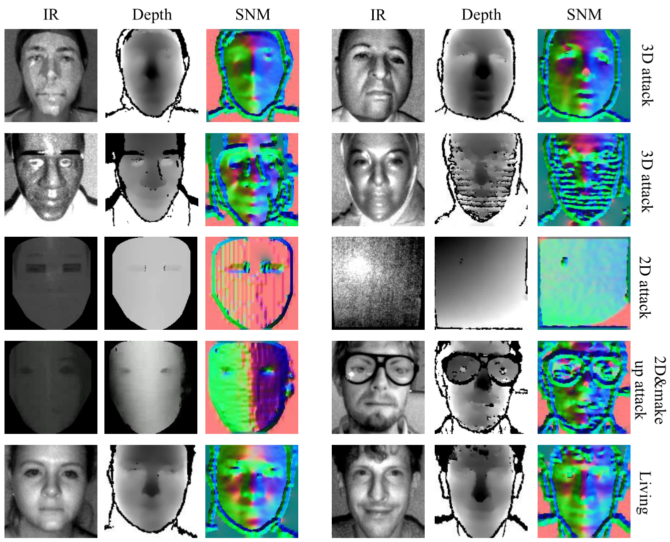


Fig. 3. Figures show living and spoofing attacks from different public databases, which include IR images, depth maps, and our proposed surface normal maps (SNMs).

This process is defined as

$$A = \frac{1}{2ks+1}\sum_{i=0}^{2ks}(P_{i,j} - \overline{PN})(P_{i,j} - \overline{PN})^{\mathrm{T}}, An = \lambda n \quad (2)$$

where $\overline{PN}$ is the mean of neighborhood points around $P_{i,j}$, ks denotes set as the neighbor radius, and A is a covariance matrix. λ, n denote the eigenvalue and eigenvector of A, respectively. Specifically, the eigenvector corresponding to the smallest eigenvalue is the normal vector $N_{i,j}$ of $P_{i,j}$. We define a set N containing normal vectors of each point in P. Each $N_{i,j}$ in N has three components representing the three axial directions. After calculating all points in the set N, we map each component set (N_x, N_y, N_z) of N to a 2D plane with the size $H \times W$ using the Gridfit [69] algorithm.

Since we define that the *x-axis* and *y-axis* coordinates of N are coordinates of the 2D plane, i.e., orthographic projection, the process of Gridfit can be expressed as

$$n_{set} = Qn_{pixel} \quad (3)$$

where the vector n_{set} comprises values of points in each component of N, the vector n_{pixel} comprises values of pixels in the 2D plane, and Q is the mapping matrix, which is determined by the interpolation method. Here, Q is similar to the orthographic projection matrix. To smooth data, we use the Springs model [70] as a regularization constraint that evaluates the surface with respect to the average value of all data. For example, we define four pixels in the 2D plane, that is, $nx_4 = [nx_{i,j}, nx_{i,j+1}, nx_{i+1,j}, nx_{i+1,j+1}]^{\mathrm{T}}$. The regularization is described as

$$B_4 nx_4 = 0, B_4 = \begin{bmatrix} -c_1 & c_1 & 0 & 0 \\ -c_1 & 0 & c_1 & 0 \\ -c_2 & 0 & 0 & c_2 \\ -c_2 & c_2 & 0 & 0 \end{bmatrix} \quad (4)$$

where we respectively set $c_1 = 1$ and $c_2 = 1/\sqrt{2}$. According to Eq. 4, we build the regularization matrix B for all pixels

Algorithm 1 Surface Normal Map Generation

Input: Depth map M_D with the size $H \times W$, based on 3D cameras

Output: Surface normal map M_{SNM} with the size $H \times W \times 3$

1: Given (i, j) is the pixel of M_D and $z_{(i,j)}$ is the depth value of M_D, where $i \in 0, 1, 2, \ldots, H$, $j \in 0, 1, 2, \ldots, W$
2: Generate input point set P, $P_{i,j} = \{i, j, z_{(i,j)}\}$
3: Initialize each normal point $N_{i,j} \in N$, $N_{i,j} = \{i, j, z_{(i,j)}\}$

4: **for all** $\{i, j, z_{(i,j)}\}) \in P$ **do**
5: Regard $P_{i,j}$ as a center point
6: Set $ks = 3$, the neighborhood points represents as $PN_{k,l}, k \in [i - ks, i + ks], l \in [j - ks, j + ks]$
7: Get the difference between the center point and $\overline{PN}$ mean of neighborhood points and build covariance matrix based on $A = \frac{1}{2ks+1}\sum_{i=0}^{2ks}(P_{i,j} - \overline{PN})(P_{i,j} - \overline{PN})^{\mathrm{T}}$
8: Return $N_{i,j}$ according to the eigenvector of A
9: **end for**
10: Divide the set N into N_x, N_y, N_z
11: Each point set is mapped into a 2D plane with the size of $H \times W$ based on Eq. 5
12: Normalize the data of each 2D plane as n_{nx}, n_{ny}, n_{nz}
13: $M_{SNM} = [n_{nx}, n_{ny}, n_{nz}]$
14: **return** M_{SNM}

in the 2D plane. Since Gridfit may be underdetermined, it is transformed into a least square problem as

$$n^* = \underset{n_{pixel}}{\arg\min} \|An_{pixel} - n_{set}\|_2 + \theta\|Bn_{pixel}\|_2 \quad (5)$$

where θ is a regularization coefficient. Based on Eq. 5, we obtain each mapped point (n_x, n_y, n_z) from each component of N. Then we normalize face data to $[0, 255]$. Finally, surface normal maps are represented by concatenating each component. As shown in Fig. 3, from the visual point of view, when compared with depth maps, the normal maps generated by our algorithm emphasize the local areas and edges with clear changes, where spoofing clues typically occur. For example, normal maps in the first row represent 3D attacks where we can easily observe the edge of masks. Compared with depth maps in the second row, normal maps of masks with different materials intend to amplify the local areas, such as the edges of the eyes region and the mouth, which is convenient for capturing spoofing information. In the fourth row, the normal map of curve prints clearly shows gradient changes in the horizontal direction. In general, these noticeable differences in edges and local regions can make it easy to distinguish between living and spoofing faces.

B. Fusion Strategy of Two-Modal Global Features

To effectively fuse information from different modalities, we introduce a Global Interaction Fusion (GIF) block based on self-attention [67]. We first establish a cross-fusion structure between the surface normal map and IR for the global complementary features. As shown in Fig 4, we denote

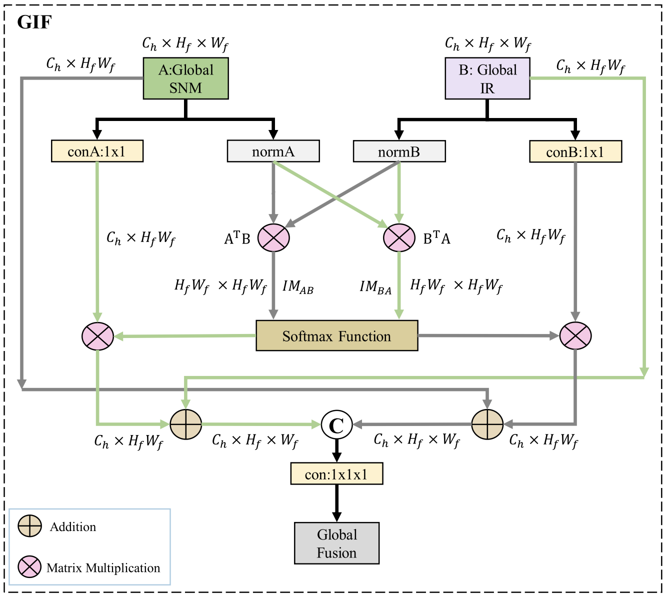


Fig. 4. The detailed structure of Global Interaction Fusion (GIF). The interaction matrices ($A^T B$ and $B^T A$) are used to enhance the SNM and IR global fusion information.

inputs of the fusion framework following channel split as $\mathbf{F}_n^g, \mathbf{F}_i^g \in R^{C_h \times H_f \times W_f}$ for normal map and IR, respectively. Then, $\mathbf{F}_n^g$ and $\mathbf{F}_i^g$ are reshaped into two matrices, defined as $\mathbf{M}_A \in R^{C_h \times N_f}$ and $\mathbf{M}_B \in R^{C_h \times N_f}$, where $N_f = H_f \times W_f$. According to the self-attention mechanism, we calculate the interaction matrices as

$$IM_{AB} = Softmax\left(\frac{\mathbf{M}_A^T \otimes \mathbf{M}_B}{norm(\mathbf{M}_A) \cdot norm(\mathbf{M}_B)}\right)$$
$$IM_{BA} = Softmax\left(\frac{\mathbf{M}_B^T \otimes \mathbf{M}_A}{norm(\mathbf{M}_B) \cdot norm(\mathbf{M}_A)}\right) \quad (6)$$

where $norm(\cdot)$ represents the L_2 normalization for the matrix, IM_{AB} means the interaction matrix from normal maps to IR and vice versa for IM_{BA}. Then we generate $\mathbf{F}_{1A}$ and $\mathbf{F}_{1B}$ via 1×1 convolution on $\mathbf{F}_n^g$ and $\mathbf{F}_i^g$ according to [71]. The fusion process can be expressed as

$$\mathbf{F}_i^f = \mathbf{F}_i^g + IM_{BA} \otimes \mathbf{F}_{1A}$$
$$\mathbf{F}_n^f = \mathbf{F}_n^g + IM_{AB} \otimes \mathbf{F}_{1B} \quad (7)$$

where $\mathbf{F}_n^f$ and $\mathbf{F}_i^f$ are the results of the fusion process, i.e., each piece of interactive information is added to each split feature map to enhance the fusion information. To obtain the final global fusion vector, we first concatenate fused two-branch global features followed by a 1×1 convolution. Subsequently, Feature Extraction Module (FEM) with an average pooling layer is utilized in the global branch. The process can be formulated as

$$F_G = f_{Avg}(f_{FEM}(f_{con}(Concat(\mathbf{F}_n^f, \mathbf{F}_i^f)))) \quad (8)$$

where f_{con}, f_{FEM}, and f_{Avg} represent the 1×1 convolution, Feature Extraction Module, and average pooling, respectively. F_G denotes the final fusion global feature.

C. Local Attention Extraction

Spoofing cues may not just exist in the global region of images. For example, it is difficult to tell whether someone wearing a silicone mask is real or not if they are standing one meter away. Correct judgments may be made based on facial details when the person is closer. In this case, the most helpful information is in subtle regions containing the edge of the mask and the corner of the eyes, nose, and mouth. Therefore, making the model more closely consider local areas can better capture discriminative features between living and spoofing faces. Our proposed methods not only consider global information as before but also pay attention to local cues by a Local Attention Extraction (LAE) module that employs learnable attention maps focusing on different contents and leverages attention maps to augment intermediate feature maps. LAE is composed of attention maps, Feature Maps Enhancement (FME), Branch Attention-guided Augmentation (BAGA), and Bilinear Attention Pooling (BAP).

1) Attention Maps: As described in Fig. 2, we first define intermediate feature maps as $\mathbf{F}_s^l$ with the size of $C_h \times H_f \times W_f$, and then leverage operations (i.e., convolution layer, batch normalization layer, and activation layer) on $\mathbf{F}_s^l$ to generate attention maps A_k, which represent different local regions of feature maps, where $A_k \in R^{M \times H_f \times W_f}$. To force the split intermediate feature maps to more closely consider subtle spoofing cues, we perform element-wise multiplication between $\mathbf{F}_s^l$ and A_k to obtain the enhanced intermediate feature maps, which can be regarded as data augmentation to improve the performance of our model.

2) Feature Map Enhancement: Previously, the role of the spoofing cues in shallow feature maps is underestimated. Distinct from the conventional methods of directly extracting feature maps with deep semantic information, to capture high-frequency cues, we design a Feature Map Enhancement block that can process features at multiple scales in Fig. 5. Specifically, we first utilize average pooling to downsample the branch feature maps $\mathbf{F}_s^l$ and resize the pooled feature maps to obtain $\mathbf{F}_{ds}^l \in R^{C_h \times H_f \times W_f}$. To highlight useful high-frequency information in the feature maps, we apply a subtraction operation between $\mathbf{F}_s^l$ and $\mathbf{F}_{ds}^l$. The whole multi-scale process of feature map enhancement is expressed as

$$\mathbf{F}_H^l = \mathbf{F}_s^l - \mathbf{F}_{ds}^l$$
$$\mathbf{F}_{out}^l = P_{rj}(Concat(f_{mcon}(\mathbf{F}_H^l))) \quad (9)$$

where $\mathbf{F}_H^l$ is the high-frequency feature map, f_{mcon} is a multi-scale convolution block with three kernels of different sizes (e.g., 3×3, 5×5 and 7×7) to enhance $\mathbf{F}_H^l$, $Concat(\cdot)$ is defined as the concatenation operation, $P_{rj}(\cdot)$ is a projection layer including a linear layer and a nonlinear activation layer, and $\mathbf{F}_{out}^l$ is the final enhanced feature map.

3) Bilinear Attention Pooling: In Fig. 2, when getting attention maps and enhanced feature maps, bilinear attention pooling (BAP) [65], [72] is used to obtain the feature matrix for each local branch. In addition, we also use BAP to generate features for loss functions, i.e., L_{cen} and L_{cos} in Fig. 5. Considering different distributions of $\mathbf{F}_H^l$, we apply the normalization to standardize $\mathbf{F}_H^l$ and leverage BAP between

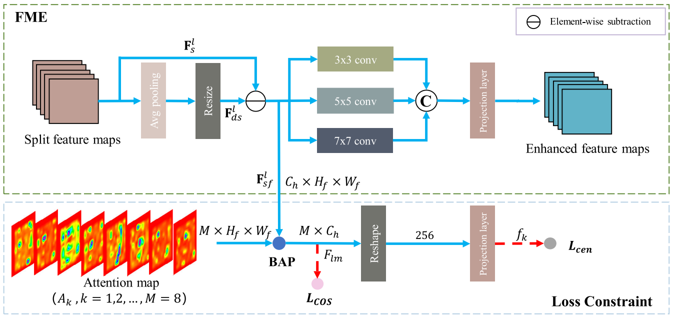


Fig. 5. The detailed architecture of Feature Map Enhancement (FME) and loss constraint makes use of a multi-scale process to enhance feature maps. The Bilinear Attention Pooling (BAP) is applied to the intermediate layer of the FME for loss functions.

standardized feature maps $\mathbf{F}^l_{sf}$ and A_k to obtain the loss feature matrix $F_{lm} \in R^{M \times C_h}$. The normalization process is written as

$$\mathbf{F}^l_{sf} = \frac{\mathbf{F}^l_H - mean(\mathbf{F}^l_H)}{std(\mathbf{F}^l_H) + \epsilon} \tag{10}$$

where $\epsilon = 1e^{-8}$, $mean(\cdot)$ and $std(\cdot)$ are functions that average different vectors and obtain the standard deviation of varying feature vectors, respectively.

4) Branch Attention-Guided Augmentation: To emphasize the local information in attention maps and effectively extract subtle spoofing cues, we adopt attention-guided augmentation in [72] to achieve data augmentation on split intermediate feature maps. For more robust feature representation, rather than using a fixed Gaussian blur kernel to obtain degraded images, we apply a random Gaussian blur kernel with two σ values picked from {3,5,7} on the feature maps during training.

D. Loss Function

As mentioned above, we not only expect that each attention map A_k represents different regions of the object but also that these regions demonstrate specific attributes during classification. To achieve this goal, we adopt the center loss [73] in Fig. 5, i.e., we penalize the variances of vectors in the loss feature matrix that belong to the same category, which signifies that various regions of the object are forced to the center of the class (living or spoofing). The center loss is represented as

$$L_{cen} = \sum_{k=0}^{M-1} \| f_k - c(y) \|_2^2,\ y \in 0, 1 \tag{11}$$

where $c(y)$ represents the center of the object, and y is the label of samples. $f_k \in R^D$ is generated by $F_{lm} \in R^{M \times C_h}$ via a reshape function and a projection layer, and $D = M \times C_h$. For values of the label, 0 means the spoofing face and 1 represents the living face. In addition, $c(y)$ can be set from zero and updated by the moving average as

$$c(y)_{k+1} = c(y)_k + \beta(f_k - c(y)_k),\ y \in 0, 1 \tag{12}$$

To learn various local features and ensure non-overlapping regions from attention maps, we use the cosine similarity to constrain vectors in loss feature matrix $F_{lm} \in R^{M \times C_h}$, where M is the number of vectors in the loss feature matrix, and C_h is the dimension of each vector. The cosine similarity loss L_{cos} is defined by

$$L_{cos} = \frac{1}{M} \sum_{0 \le m,n \le M, m \ne n} \left(\frac{v_m \cdot v_n}{\|v_m\| \|v_n\|} \right)^2 \tag{13}$$

where v_m and v_n are different feature vectors of the feature matrix. Therein, we enlarge the intraclass distance to find helpful information about spoofing cues in different areas of the image.

Although our framework produces two classification outputs from two stages, we only add a binary cross-entropy loss to the second stage output. The classification loss L_c is defined as

$$L_c = \frac{1}{Num} \sum_i \left[y_i log(o2_i) + (1 - y_i) log(1 - o2_i) \right] \tag{14}$$

where Num is the number of samples, y_i denotes the label of the i_{th} sample, and $o2_i$ belongs to the i_{th} output of the second stage, which is the predicted value of the i_{th} sample.

In summary, the supervisions mentioned above are employed jointly to leverage global and local fusion information fully. The overall loss $L_{overall}$ is defined as

$$L_{overall} = \alpha L_{cen} + \beta L_{cos} + L_c \tag{15}$$

where α and β are the balancing weights for the three terms. According to the results of Table IV, we set $\alpha = \beta = 1$ in our experiments.

IV. Experiments

In this section, we conduct a series of experiments to demonstrate the effectiveness of our proposed method. First, we visually and quantitatively indicate the superiority of our proposed surface normal maps. Then we explore the effectiveness of each proposed block and setting. Finally, we compare our framework with the state-of-the-art multimodal methods on three public databases.

A. Multimodal Database

Three databases, including CASIA-SURF [24], CeFA [26], and WMCA [27], are utilized to assess the effectiveness of our proposed FAS method. CASIA-SURF is a large-scale multi-modal database consisting of 1,000 subjects with 21000 videos corresponding to three modalities (i.e., RGB, IR, and depth) and provides an intra-testing protocol. WMCA has a wide variety of 2D and 3D attacks, which provides two protocols, i.e., "seen" and "unseen". CeFA is a cross-ethnicity database, covering three ethnicities (i.e., African, East Asian, and Central Asian) and three modalities (i.e., RGB, IR, and Depth), and provides five protocols according to different ethnicities and presentation attack instruments (PAI).

TABLE I

EFFECT OF VARIOUS MODALITIES ON CASIA-SURF, WMCA, AND CEFA. A LARGE TPR (%) AND A LOWER ACER (%) INDICATE BETTER PERFORMANCE. NOTE THAT RESNET18 IS THE BASELINE FOR COMPARISON. BEST RESULTS ARE BOLDED

Database	Protocol	Modality	APCER(%)	BPCER(%)	ACER(%)	TPR(%)		
						@FPR=10^{-2}	@FPR=10^{-3}	@FPR=10^{-4}
CASIA-SURF	intra-testing	IR	2.53	3.27	2.90	90.52	54.86	20.68
		RGB	12.72	12.28	12.50	50.28	22.00	12.28
		Depth	1.58	1.54	1.56	97.69	77.60	52.74
		SNM	**0.74**	**0.49**	**0.62**	**99.63**	**97.21**	**89.60**
WMCA	seen	IR	0.57	0.40	0.48	99.91	98.81	96.98
		RGB	2.24	1.45	1.85	94.31	50.60	48.25
		Depth	**0.01**	1.43	0.72	99.95	99.33	**98.69**
		SNM	0.17	**0.02**	**0.09**	**100.00**	**99.69**	92.31
CeFA	4@1	IR	**0.1**	9.25	4.68	97.00	89.50	86.40
		RGB	39.57	10.50	25.13	3.25	-	-
		Depth	63.98	**0.25**	32.07	1.25	-	-
		SNM	0.36	1.25	**0.81**	**99.75**	**98.50**	**97.75**
	4@2	IR	8.74	7.00	7.78	24.75	5.25	-
		RGB	51.72	38.00	44.86	-	-	-
		Depth	65.19	**4.75**	34.97	-	-	-
		SNM	**0.00**	11.25	**5.63**	**98.25**	**94.25**	-
	4@3	IR	11.50	**2.00**	**6.75**	10.25	-	-
		RGB	36.78	37	36.89	-	-	-
		Depth	64	20.25	42.12	-	-	-
		SNM	**0.00**	14.00	7.00	**92.25**	**88.5**	**86.25**

B. Experimental Setting & Metrics

To validate the advantage of our proposed normal maps, we first conduct baseline experiments performed on various modalities. The selected baselines are reproducible and have open-source implementation. Thus, we choose ResNet18 [66] with cross-entropy loss as a baseline for testing single-modal information. Specifically, we remove the RGB stream of multimodel networks for fair comparisons with prior works. In addition, we choose the "seen" and "unseen" protocols [27], [33] for WMCA and select the most challenging protocol, *Protocol 4* (cross-ethnicity & PAI) for CeFA.

In the experiments, Attack Presentation Classification Error Rate (APCER), BonaFide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER) [74] are utilized as evaluation metrics via given decision threshold on CASIA-SURF, WMCA, and CeFA. For the "unseen" WMCA protocol, the decision threshold is determined by BPCER=1% on the dev set. For the "seen" protocol of WMCA, CASIA-SURF, and CeFA, the decision threshold can be calculated according to the Equal Error Rate (EER) on the dev set. Apart from the error metrics, following the metrics of face recognition, the True Positive Rate (TPR) and False Positive Rate (FPR) are also utilized to evaluate the results. In particular, TPR@FPR=10^{-2}, TPR@FPR=10^{-3} and TPR@FPR=10^{-4} are adopted to show the effectiveness of our proposed method.

C. Implementation Details

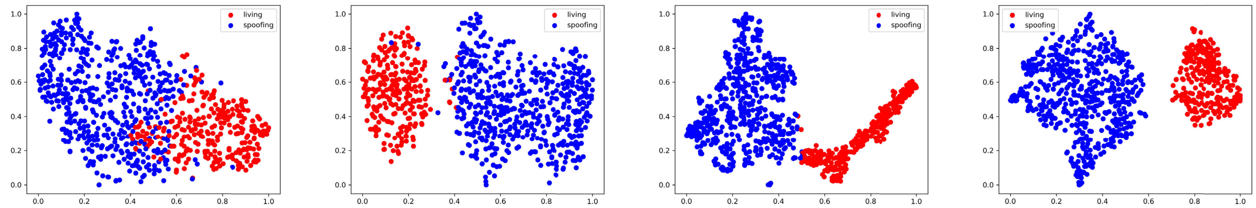
Following previous works [25], [26], we resize the cropped face region to 112×112 and use data augmentation, including random rotation, resizing, flipping, cropping, and color distortion. All experiments are trained for 60 epochs via the SGD optimizer with momentum as 0.9 on one RTX3090 GPU. The learning rate is set to 0.001 for training the network decreased by CosineLR. The weight decay and batch size are set to 0.0005 and 128, respectively.

D. Baseline Model Evaluation For Single-Modal Inputs

In this subsection, we analyze the discrimination and robustness of extracted spoofing features via the baseline. Note that all training processes follow the provided protocols according to Section IV-B.

As shown in Table I, we find that RGB images exhibit undesirable performance on CASIA-SURF. Since the database captures scenes under different environmental conditions (e.g., low-lighting environment), some RGB images may provide little valuable information. In contrast, IR sensors have stronger adaptability than RGB cameras in tackling different environmental constraints. In particular, the ACER is reduced to 2.90%, which shows that IR images can capture helpful cues for FAS due to their imaging properties. For depth maps, the results of TPR@FPR $= [10^{-2}, 10^{-3}, 10^{-4}]$ are raised to 97.69%, 77.60%, and 52.74%, respectively. The reason behind it is that the database mainly contains 2D attacks with apparent shape discrimination compared with living faces. The depth map contains spatial information, which gives it an advantage in judging 2D attacks. Significantly, surface normal maps with more discriminative information perform best among these four modalities.

Regarding WMCA, no new attack types appear in the $test$ set, which reduces the challenge of the task. In Table I, the best results are achieved by using normal maps, reaching 0.09%, 100%, 99.69% for ACER, TPR@FPR $= [10^{-2}, 10^{-3}]$. In particular, depth and IR modalities perform better than normal maps for TPR@FPR $= 10^{-4}$. Because there are



(a) RGB (b) IR (c) Depth (d) SNM

Fig. 6. The t-SNE [75] visualization of different modalities on CASIA-SURF.

more 3D attacks in WMCA, IR modality show remarkable results on error rate. Similar to the results on CASIA-SURF, RGB modality is poorly evaluated. The results of TPR@FPR $= [10^{-3}, 10^{-4}]$ decrease rapidly to 50.60% and 48.25%, respectively.

To further prove the effectiveness of normal maps, we select the most challenging protocol in CeFA, which simultaneously contains cross-ethnicity and cross-PAI. Moreover, to increase the challenge, 3D attacks are only included in the *test* set. All subprotocols achieve poor performance, demonstrating the limitation of the single modality. Depth modality shows undesirable performance, which is attributed to unseen attacks appearing in the *test* set, resulting in models being unable to accurately acquire discriminative information between living and spoofing faces. The results of RGB images are also poor, further demonstrating the above shortcomings of RGB modality for FAS. In contrast, using IR images, the ACERs can be decreased to 4.68%, 7.78%, and 6.75% for 4@1, 4@2, 4@3. Significantly, normal maps achieve remarkable results in three subprotocols. Although the ACER of normal maps in 4@3 is less than that of IR images, other metrics have reached the best performance, and TPR@FPR $= [10^{-2}, 10^{-3}, 10^{-4}]$ shows higher values than other modalities.

In addition, we design visualization experiments on CASIA-SURF to show the feature spaces of different modalities. Specifically, we randomly select 800 samples in the testing set. Then, we extract high-dimensional features from different modalities by the baseline. Finally, t-SNE [75] is adopted for dimension reduction to obtain 2D visualization maps. As shown in Fig. 6, the features of RGB are confusing. Living and spoofing faces cannot be easily distinguished, and the intra-class space is not compact enough. For the IR and depth, there are clear boundaries between the living and spoofing faces. However, the intra-class interval of living faces in depth and spoofing faces in IR are slightly large. The result of our normal map indicates larger inter-class and smaller intra-class distances, which is essential for classification. All these demonstrate the effectiveness of the normal maps.

E. Ablation Study

To further investigate the advantage of our proposed framework and the contributions of individual model blocks, we experiment on some incomplete models that control for different steps in the algorithmic flow. Ablation studies are conducted on CASIA-SURF and WMCA, which report both quantitative and qualitative results.

TABLE II

EVALUATION RESULTS OF EACH BLOCK ON CASIA-SURF. NOTE THAT "S", "G", "F", AND "BA" REPRESENT SURFACE NORMAL MAPS, GLOBAL INTERACTION FUSION, FEATURE MAP ENHANCEMENT, AND BRANCH ATTENTION-GUIDED AUGMENTATION, RESPECTIVELY. BEST RESULTS ARE **BOLDED**

Model	APCER (%)	BPCER (%)	ACER (%)	TPR (%) @FPR= 10^{-2}	TPR (%) @FPR=10^{-3}
Model1 (baseline)	0.41	1.74	1.08	99.20	95.22
Model2 (S)	0.55	1.21	0.88	99.50	60.38
Model3 (S&G)	0.41	1.22	0.81	99.64	93.57
Model4 (S&G&BA)	0.32	**0.31**	0.31	99.91	98.98
Model5 (S&G&BA&F)	**0.08**	**0.31**	**0.20**	**99.94**	**99.73**

1) Effect of Each Block: As shown in Table II, different combinations of the blocks in our framework yield six variants. Specifically, (1) Model 1: the fusion baseline based on residual blocks with depth and IR modalities. (2) Model 2: Using the normal map instead of depth. (3) Model 3: Adding Global Interaction Fusion (GIF) block based on Model 2. (4) Model 4: Developing the Branch Attention-guided Augmentation (BAGA) block based on Model 3. (5) Model 5: Adding the Feature Map Enhancement (FME) block.

Comparing the results of Model 1 and Model 2, we find that the normal map improves the performance of multimodal FAS, especially for ACER and TPR@FPR $= 10^{-2}$. Rather than using summation, the GIF block enhances the representation ability of fusion features. At the same time, Model 3 achieves comparable performance compared with Model 2. When we add local cues to the task, Model 4 shows remarkable improvements, which indicates that the local cues are beneficial to multimodal FAS. Through the comparison of Model 4 and Model 5, we can verify the effectiveness of the FME block. In general, our proposed framework achieves the best results due to the contributions of each block mentioned above.

2) Analysis of Loss Functions: To confirm the importance of each loss function, we first define L_{bce} as a basic loss function [18], [50], and then we make the framework with or without L_{cos} and L_{cen} to show the significance of different loss functions. In Table III, only using L_{bce} yields poor results. To ensure that the model focuses on various local regions of attention maps, we apply L_{cos} to the first stage of the framework, which achieves a significant improvement in TPR@FPR $= 10^{-3}$. Moreover, we also observe that adding L_{cen} improves the performance compared with only the basic loss function. Finally, our proposed joint loss functions can achieve the best results.

TABLE III

Analysis of Loss Functions on CASIA-SRUF. A Higher TPR (%) and a Lower ACER (%) Indicate Better Performance. Best Results Are **Bolded**

L_{bce}	L_{cos}	L_{cen}	APCER (%)	BPCER (%)	ACER (%)	TPR (%) @FPR= 10^{-2}	TPR (%) @FP$R = 10^{-3}$
✓			0.62	1.23	0.93	99.47	87.81
✓	✓		0.66	0.93	0.8	99.48	94.63
✓		✓	0.33	0.56	0.44	99.85	97.07
✓	✓	✓	**0.08**	**0.31**	**0.20**	**99.94**	**99.73**

TABLE IV

Sensitivity to Balancing Weights of Loss Functions on CASIA-SURF. A Higher TPR (%) and a Lower ACER (%) Indicate Better Performance. Best Results Are **Bolded**

α	β	APCER (%)	BPCER (%)	ACER (%)	TPR (%) @FPR= 10^{-2}	TPR (%) @FP$R = 10^{-3}$
$\alpha = 0.5$	$\beta = 1$	0.60	0.64	0.62	99.61	96.49
$\alpha = 1$	$\beta = 0.5$	0.55	0.76	0.65	99.76	93.50
$\alpha = 0.5$	$\beta = 0.5$	0.58	1.23	0.90	99.32	88.42
$\alpha = 1$	$\beta = 1$	**0.08**	**0.31**	**0.20**	**99.94**	**99.73**

3) Sensitivity to Balancing Weights of Loss Functions: We conduct experiments to study the sensitivity of different weights of loss functions. As shown in Table IV, we observe that there is a noticeable performance degradation when values of α and β are reduced, especially $\alpha = 0.5, \beta = 0.5$ reducing below 90% for TPR@FPR $= 10^{-3}$. According to the best results, we set $\alpha = \beta = 1$ in the following experiments.

4) The Number of Attention Maps: We conduct experiments to analyze the number of attention maps. Specifically, we select different numbers of attention maps (we denote the number as M), training the model on the CASIA-SURF database. As shown in Table V, $M = 4$ shows a sharp decline for TPR@FPR= $[10^{-3}, 10^{-4}]$. Although the ACER and TPR@FPR $= 10^{-2}$ of $M = 6$ is close to $M = 4$, the performance for TPR@FPR= $[10^{-3}, 10^{-4}]$ is improved to 97.39% and 80.52%. Significantly, $M = 8$ reaches the best results. When further increasing the M again, we note that the evaluation metrics decrease seriously. Therefore, we set the value of M as 8.

5) Kernel Size of Branch Attention-Guided Augmentation: In the Branch Attention-guided Augmentation (BAGA) block, we apply Gaussian kernels on attention maps to produce different local cues. Two kinds of Gaussian kernels, i.e., anisotropic and isotropic kernels, are widely used in image processing. We randomly select σ from various σ sets to build anisotropic and isotropic kernels. Table VI shows that fully using both anisotropic and isotropic kernels generates remarkable results compared with only isotropic kernels, especially for the TPR@FPR $= 10^{-4}$.

6) Visualization: As shown in Fig. 7, distinct from "w/o GIF", we observe that results trained with GIF have large values in face areas, and the difference between these values is relatively small, which indicates that the model with GIF pays more attention to global features of face regions. In Fig. 8, for each IR image, it is observable that the attention maps of the first line always focus on the same local region, which deviates from our original intention. Each attention map of the second line does not overlap, which is convenient for capturing more subtle cues. Moreover, the attention maps focus on different regions according to types of spoofing attacks. For example, attention maps for 2D attacks are interested in areas with sharp pixel changes, while often focus on areas of the eyes, nose, and mouth for 3D attacks. Compared with the images in Fig. 7 and Fig. 8, we find that the feature maps in the global branch focus on the whole faces due to the small color span. In contrast, attention maps in the local branch concern various subtle regions because of the large color span. The above analysis demonstrates that our model simultaneously focuses on local and global information and further verifies the effectiveness of GIF and L_{cos}.

TABLE V

Evaluation Results of Different Attention Maps on CASIA-SURF. A Higher TPR (%) and a Lower ACER (%) Indicate Better Performance. Best Results Are **Bolded**

M	APCER (%)	BPCER (%)	ACER (%)	TPR(%) @FPR=10^{-2}	TPR(%) @FPR=10^{-3}	TPR(%) @FPR=10^{-4}
4	0.54	0.45	0.50	99.89	76.45	6.40
6	0.41	0.63	0.52	99.79	97.39	80.52
8	**0.08**	**0.31**	**0.20**	**99.94**	**99.73**	**97.12**
10	0.73	1.12	0.92	99.29	85.04	15.86

TABLE VI

Evaluation Results of Different σ Sets on CASIA-SURF Database. A Higher TPR@FPR=10^{-4} (%) and a Lower ACER (%) Indicate Better Performance. Best Results Are **Bolded**

σ sets	APCER(%)	BPCER(%)	ACER(%)	TPR@FPR=10^{-4} (%)
[3,3]	0.52	0.6	0.56	59.71
[5,5]	0.37	0.34	0.35	84.69
[7,7]	0.25	**0.26**	0.26	94.38
[3,5,7]	**0.08**	0.31	**0.20**	**97.12**

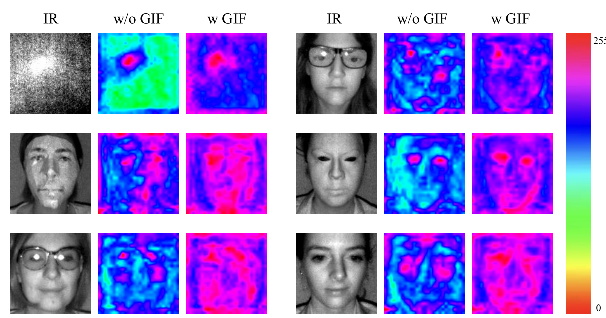


Fig. 7. The visualization of feature maps in the global branch trained without a Global Interaction Fusion block (w/o GIF) and with a Global Interaction Fusion block (w GIF) is implemented on WMCA. "w GIF" shows larger values with slight differences concentrated in the whole face region, while "w/o GIF" depicts larger values distributed across the entire image, and it is difficult to distinguish the face area.

F. Multi-Stream FAS for Multimodal Database

According to the protocols for different databases, these databases are split into three subsets, i.e., *train* set, *validation* set, and *test* set. In this experiment, we keep the

TABLE VII

Evaluation Results on the Intra-Testing Protocol of CASIA-SURF. A Higher TPR (%) and a Lower ACER (%) indicate better Performance. Note That R, I, D, and S Are Shot for RGB, IR, Depth, and SNM, Respectively. Best Results Are **Bolded**

Method	Inputs	APCER (%)	BPCER (%)	ACER (%)	TPR (%)			Param (M)
					@FPR=10-2	@FPR=10-3	@FPR=10-4	
MS-SEF [24]	I&D	0.42	0.8	0.61	99.65	96.03	77.27	22.52
MC-PixBis [76]	I&D	7.63	1.61	4.62	87.84	-	-	-
FaceBagNet [32]	I&D	0.59	0.22	0.40	**99.94**	94.67	74.79	41.24
PipeNet [31]	I&D	2.08	2.45	2.26	93.01	72.49	55.99	63.28
SCA [63]	I&D	1.64	2.66	2.15	94.56	16.3	-	7.49
MM-CDCN [28]	I&D	1.25	3.13	2.19	96.34	82.67	39.39	5.08
PSMM [26]	I&D	0.1	0.78	0.44	99.87	99.16	95.78	33.39
ViT-S [62]	I&D	0.44	1.17	0.81	99.50	95.17	57.89	21.67
AMT [77]	I&D	0.20	1.73	0.96	99.66	-	-	-
ViT-Base_CA [4]	I&D	1.33	1.67	1.50	97.87	89.11	13.3	91.11
Ours	I&S	0.08	0.31	**0.20**	**99.94**	**99.73**	**97.12**	**4.03**

TABLE VIII

Evaluation Results on WMCA. ACER Values Are reported. A Lower ACER (%) Indicates Better performance. Note That I, D, and S Are Shot for IR, Depth, and SNM, Respectively. Best Results Are **Bolded**

Method	**Inputs**	**Seen**	**Unseen**							
			Flexiblemask	**Replay**	**Fakehead**	**Print**	**Glasses**	**Papermask**	**Rigidmask**	**mean±std**
MCNN [27]	I&D	0.88	30.51	0.05	1.14	0.00	50.06	0.70	15.45	13.99±19.61
PipeNet [31]	I&D	1.14	18.24	1.06	1.45	1.17	50.12	0.58	0.82	10.49±18.62
FaceBagNet [32]	I&D	0.25	21.40	0.52	23.08	0.00	50.00	0.29	4.61	14.27±18.65
MS-SEF [24]	I&D	0.52	17.70	2.33	0.52	0.07	50.20	0.59	2.72	10.59±18.54
PSMM [26]	I&D	0.46	16.95	1.0	28.46	1.52	50.00	0.58	0.82	14.19±19.11
MM-CDCN [28]	I&D	0.02	33.16	1.27	10.14	0.28	46.54	0.37	1.47	13.32±18.84
SCA [63]	I&D	0.88	21.19	0.54	47.54	0.01	49.82	2.37	2.68	17.74±22.37
ViT-S [62]	I&D	1.01	21.38	0.39	3.61	0.84	50.00	0.01	4.2	11.49±18.55
ViT-Base_CA [4]	I&D	1.20	36.70	0.09	15.31	0.55	50.49	4.55	20.79	18.35±19.26
Ours	I&S	**0.015**	18.07	1.43	0.44	0.69	50.00	0.39	1.37	**10.34±18.65**

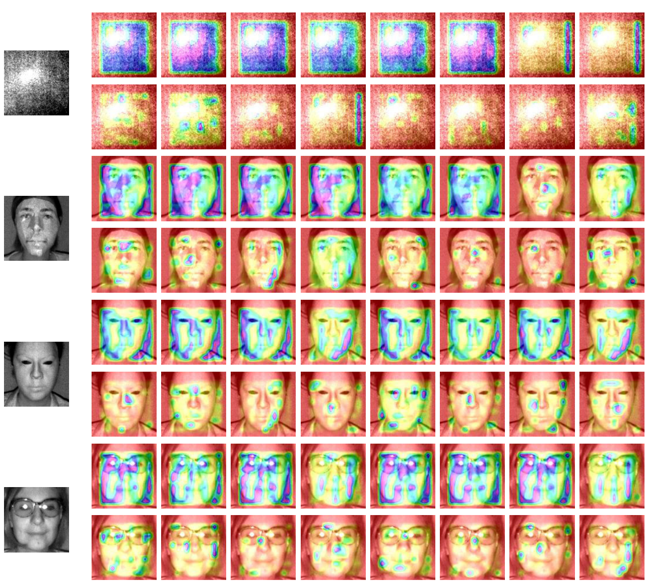


Fig. 8. The visualization of attention maps trained without L_{cos} (w/o L_{cos}) and with L_{cos} (w L_{cos}) are implemented on WMCA. The first line of each IR image (w/o L_{cos}) demonstrates that attention maps contain many overlapping areas, while the second line (w L_{cos}) focuses on different local regions. Note that the color range is consistent with Fig. 7.

same experimental settings and use prior multimodal works for evaluation to explore the advantages of our approach.

1) Results of CASIA-SURF: In Table VII, by contrast with multimodal methods, i.e., MS-SEF [24], MC-PixBis [76], MM-CDCN [28], PSMM [26], and recent AMT [77], our proposed approach achieves the best performance, which proves its generalization capability facing unseen spoofing attacks. Note that FaceBagNet [32] and PipeNet [31] mainly focus on local cues using random patches of inputs. SCA [63], ViT-S [62], and ViT-Base_CA [4] are attention-based methods, which employ channel and spatial attention, self-attention, and cross-attention mechanisms, respectively. Compared with these methods, we can see that the remarkable accuracy of ACER is 0.2%, showing that our model can capture meaningful spoofing cues with both local and global fusion information via the attention mechanism. Moreover, we can also observe that our proposed method has fewer parameters than other methods, which is more suitable for practical application.

2) Results of WMCA: For the WMCA database, we adopt the "seen" and "unseen" protocols. The quantitative results are depicted in Table VIII. Since the *test* set covers all the attack types of the *train* set on the "seen" protocol, all results achieve relatively favorable performance. Our approach achieves the best performance on "seen" and "unseen" protocols, which proves its local and global cues to be significant for multimodal FAS. In addition, PipeNet [31] using random patches

TABLE IX

EVALUATION RESULTS ON *Protocol 4* OF CEFA. A LOWER ACER INDICATES BETTER PERFORMANCE. BEST RESULTS ARE **BOLDED**

Method	Protocol	APCER (%)	BPCER (%)	ACER (%)
MS-SEF [24]	4@1	2.97	4.00	3.48
	4@2	9.94	2.25	6.09
	4@3	6.04	2.00	4.02
	Avg±Std	4.97±4.33	2.75±1.09	4.53±1.38
PipeNet [31]	4@1	3.85	0.50	2.18
	4@2	2.81	2.00	**2.40**
	4@3	10.46	5.50	7.98
	Avg±Std	5.71±4.15	**2.67±2.57**	4.19±3.29
MM-CDCN [28]	4@1	1.40	16.00	8.7
	4@2	4.11	9.00	6.56
	4@3	4.58	2.50	3.54
	Avg±Std	3.36±1.72	9.17±6.75	6.27±2.59
PSMM [26]	4@1	0.36	1.75	1.06
	4@2	6.19	6.25	6.22
	4@3	0.88	11.5	6.19
	Avg±Std	2.47±3.23	6.50±4.88	4.49±2.97
ViT-S [62]	4@1	29.19	3.25	16.22
	4@2	37.93	4.5	21.21
	4@3	10.67	16.25	13.46
	Avg±Std	25.93±13.92	8.00±7.17	16.96±3.93
ViT-Base_CA [4]	4@1	15.30	3.25	9.27
	4@2	12.38	4.75	8.57
	4@3	4.89	9.75	7.32
	Avg±Std	10.86±5.37	5.92±3.40	8.36±1.04
Ours	4@1	0.00	1.50	**0.75**
	4@2	3.17	7.25	5.21
	4@3	0.31	3.25	**1.78**
	Avg±Std	**1.16±1.75**	4.00±2.95	**2.58±2.33**

produces undesirable results on the "seen" protocol, which is attributed to random patches and the designed framework losing some important semantic information for the FAS task. The recent attention-based method, ViT-Base_CA [4], also shows poor performance in both "seen" and "unseen" protocols. One reason is attributed to the removal of the RGB stream, which has a great impact on the attention-based fusion module. The other reason is that larger parameters of ViT-Base_CA lead to overfitting on "unseen" protocols.

3) Results of CeFA: For the CeFA database, we choose the most challenging *protocol 4* for experiments. Table IX shows methods on CeFA perform even worse than other databases. One crucial reason is adding unseen 3D attacks in the *test* set, which increases the difficulty of detection. Instead of promising results on the WMCA database, MM-CDCN [28] attains poor performance for ACER. This phenomenon is attributed to the lack of data in RGB samples, which affects the generalization of the MM-CDCN model on cross-ethnicity & cross-PAI data. It is worth noting that PipeNet [31] achieves the best value on the 4@2 protocol, which verifies the importance of local cues for FAS. However, PipeNet cannot show robust results on the other two subprotocols, indicating that depending only on local features is not enough. Although ViT-S [62] and ViT-Base_CA [4] use different attention mechanisms to focus on important regions, they achieve poor performance on CeFA. We argue that further subdividing these regions into local and global areas is useful for fusion. Different from the aforementioned methods, our proposed method considering both local and global fusion features yields the best results, which further verifies that our approach is practical on more challenging databases and explains the significance of our proposed network structure.

V. CONCLUSION

We have presented a novel attention-aware dual-stream framework based on 3D cameras that enables the learning of data fusion representations from both local and global cues. Specifically, the framework employs surface normal maps (SNMs) rather than depth maps to realize powerful representations for FAS. Then, a Global Interaction Fusion (GIF) block is deployed to fuse SNM and IR images for global information effectively. Simultaneously, a two-stage feedback Local Attention Extraction (LAE) module is designed for each modality to extract discriminative local cues by learnable attention maps. Moreover, a joint loss function is adopted to ensure the accuracy of the classification. Extensive experiments on main benchmark databases demonstrate that our method can attain state-of-the-art performance.

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